Hariprashad Ravikumar

PhD Candidate specializing in Physics-Based Modeling, Predictive Analytics, and AI for Science

Website: hariprashad-ravikumar.github.io
Email: hari1729@nmsu.edu
LinkedIn: linkedin.com/in/hariprashad-ravikumar
Phone: +1 575-249-9610
GitHub: github.com/Hariprashad-Ravikumar
Location: San Jose, CA

Summary

PhD physicist with 5+ years in HPC, ML for science, and physics-based simulation. Shipped production tools (Dash, CI/CD, Kubernetes, PyPI) at Western Digital. Seeking full-time roles in data science, ML engineering, or computational physics.

Experience

1. **HAMR Modeling and Simulation Intern**, Western Digital, San Jose, CA *(May 2026 – Aug 2026)*
   - Built and shipped a full-stack physics simulation web app (Python, Dash, Plotly) that lets engineers run experiments computationally instead of on physical hardware; used daily in production by 40+ senior R&D engineers and subject-matter experts across the company's US and Japan sites
   - Added real-time parameter sweeps, side-by-side run comparison, and one-click auto-export to PowerPoint/Excel; built client-side state management with IndexedDB API for persistent, cross-tab data handoff
   - Packaged the tool as a modular, PyPI-installable Python library with a clean API; deployed on Kubernetes via a Jenkins CI/CD pipeline and integrated it into a broader internal engineering platform alongside related analysis tools
   - Derived a closed-form analytical model of magnetic grain switching behavior from first principles, combining thermal-activation and coherent-rotation theory to predict switching time, signal noise, and error probability across repeated write cycles; validated against Monte Carlo simulation and experimental lab data, with core findings in preparation for peer-reviewed publication
   - Applied statistical modeling and curve fitting to identify causal drivers of storage performance, letting engineers predict and reduce data-integrity errors computationally, removing a hardware-test step from the R&D loop and cutting testing costs
   - Presented results at PhD Expo 2026 (50+ attendees) and multiple internal engineering reviews (30+ attendees each), including a cross-site review with the company's Japan R&D team
2. **Graduate Research Assistant**, NM State University, Las Cruces, NM *(Aug 2021 – Present)*

   PhD Project: Lattice Quantum Chromodynamics & Machine Learning Approaches
   - Generated 30,000+ high-fidelity synthetic data points by solving Partial Differential Equations with large-scale Hybrid Monte Carlo simulations (Markov Chain stochastic process) and built an end-to-end AI for Science pipeline to model the underlying physics, achieving over 98% predictive accuracy with symbolic regression machine learning.
   - Engineered parallelized Bayesian inference pipelines across multi-CPU architectures to fit non-linear, high-dimensional models; utilized automatic differentiation and full covariance matrices to ensure exact error propagation and numerical stability in multi-stage workflows (jackknife resampling)

Independent Collaborations

1. **Los Alamos National Laboratory** - Lattice Quantum Chromodynamics *(May 2024 – Present)*
   - Accelerated multi-terabyte scientific calculations by developing and optimizing parallelized C++ CUDA kernels for GPU-accelerated HPC clusters (NERSC Perlmutter), significantly reducing runtime for large-scale Hamiltonian Monte Carlo simulations.
   - Managed and executed 75,000+ CPU/GPU compute hours by designing and deploying custom SLURM workflows for large-scale job orchestration
2. **North Carolina State University** - Mathematical Physics *(Dec 2020 - Present)*
   - Implemented and managed Mathematica symbolic computation workflows on HPC clusters to analyze complex algebraic structures and symmetry constraints.

Technical Projects

1. **Cost-Aware Multi-Agent LLM Router** GitHub | Live Demo
   - Cut LLM inference cost 93.9% versus always using the top-tier model, with no statistically distinguishable accuracy loss on a 50-prompt disjoint holdout, by designing a calibrated confidence router (logistic regression over logprob uncertainty, self-consistency dispersion, and embedding-distance features) that escalates only the prompts likely to need a costlier model
   - Validated router confidence for honesty, not just raw accuracy, using Expected Calibration Error and Brier score, then deployed it as a production FastAPI service with Postgres decision logging, Redis caching, and an async circuit breaker on Google Cloud Run
2. **AI-DataScience-Lab: Full-Stack Data Product** GitHub | Live App
   - Prototyped and deployed a scalable full-stack data tool (Python, Flask, Azure) to deliver on-demand data-driven insights and actionable recommendations, integrating an OpenAI API to demonstrate automated data storytelling and data interpretation.
   - Implemented an end-to-end statistical analysis pipeline using Python (Pandas) for data cleaning, statistical modeling (Scikit-learn) for regression, and Matplotlib for data visualization.

Selected Publications

Google Scholar

1. C.-R. Ji and **H. Ravikumar**, *"Interpolating conformal algebra in (1+1) dimensions,"* **Physical Review D** 113, 096018 (May 2026). DOI: 10.1103/prlq-4j1l.

Technical Skills

Programming	Python, C++, CUDA, Bash, SQL, Lua, HTML/CSS, YAML
ML & APIs	PyTorch, TensorFlow, Scikit-learn, Pandas, Dash, Plotly, Flask, FastAPI, Numba
Cloud & MLOps	Azure, AWS (S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD, Git, SLURM
Methods & HPC	Parallel Computing (GPU, MPI), Numerical Methods (PDEs, Monte Carlo, Regression)

Education

PhD in Physics, New Mexico State University, USA *Aug 2021 – Dec 2026 (expected)*
MS in Physics, New Mexico State University, USA *Aug 2021 – May 2024*

Certifications

- Fundamentals of Accelerated Computing with CUDA Python by NVIDIA
- Advanced Learning Algorithms by DeepLearning.AI
- Supervised Machine Learning: Regression and Classification by DeepLearning.AI
- Google Advanced Data Analytics Professional Certificate

Awards

- **2025 NMC Collaboration Grant**, awarded by the New Mexico Consortium to conduct my independent research project in collaboration with scientists at Los Alamos National Laboratory
- **2023 George and Barbara Goedecke Physics Excellence Fund Scholarship**, awarded by the NMSU Physics Department

Leadership

- **Vice President**, Physics Graduate Student Organization (NMSU) *Aug 2025 – Present*
  Organized professional development events and served as the primary liaison between 40+ graduate students and faculty.

Relevant Graduate Coursework

- Quantum Computing, Advanced Computational Physics, Statistical Mechanics, Electromagnetic Theory I & II